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LITHIUM RECOVERY USING AQUEOUS SOURCES

Abstract

Described herein are methods of recovering lithium from dilute lithium sources. The methods include extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate, performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. At least one of the concentration operations includes a membrane separation operation having a plurality of reactors in series each having a semi-permeable membrane, such as a counter-flow reverse osmosis operation. Methods may also include generating a low TDS stream as a permeate from any of the one or more concentration operations, wherein the low TDS stream is recycled or used as fresh water.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application claims benefit of U.S. Provisional Patent Application Ser. No. 63/364, 142 filed May 4, 2022, which is entirely incorporated herein by reference, and from Application Ser. No. 63/374,441 filed Sep. 2, 2022, which is entirely incorporated herein by reference, and from International Application No. PCT/US2022/051500, filed Dec. 1, 2022, which is entirely incorporated herein by reference.

FIELD

[0002] This patent application describes methods and apparatus for lithium recovery from aqueous sources. Specifically, effective processes for concentrating and recovering lithium from dilute sources are described.

BACKGROUND

[0003] Lithium is a key element in energy storage. Electrical storage devices, such as batteries, supercapacitors, and other devices commonly use lithium to mediate the storage and release of chemical potential energy as electrical current. As demand for renewable, but non-transportable, energy sources such as solar and wind energy grows, demand for technologies to store energy generated using such sources also grows.

[0004] According to the United States Geological Survey, global reserves of lithium total 22 million tons (metric) of lithium content, with Chile, Australia, Argentina, and China accounting for about 85% of global reserves. U.S. Geological Survey, Mineral Commodity Summaries, January 2022. According to S&P Global Market Intelligence, lithium supply is forecast to be 636 kT LCE in 2022, up from 497 kT in 2021. Global consumption was estimated at 64 kT in 2021, putting current lithium supplies in deficit. Global consumption and is expected to reach 2 MTa by 2030 for an average annual growth in demand of approximately 13.5%. Supply is currently forecast to run behind demand, and lithium prices currently outstrip even the most optimistic forecasts. While lithium prices are quite volatile as the global market develops, lithium prices are expected to remain high through 2030. The incentive for more lithium production could not be clearer.

[0005] Lithium extraction from brine has become a favored method of lithium recovery. Heretofore, most development has been focused on brine sources with relatively high concentrations of lithium, but other more dilute sources are also plentiful. Effective and efficient processes for recovering lithium from dilute sources are needed.

SUMMARY

[0006] The disclosure relates to a method of recovering lithium from a lithium source, comprising

extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; concentrating the lithium intermediate in an impurity preparation stage to yield an impurity stage feed; and treating the impurity stage feed in an impurity stage to remove impurities and to form a purified lithium stream.

[0007] The disclosure also relates to a method of recovering lithium from an aqueous lithium source, comprising separating lithium using a lithium selective electrochemical separation process; and concentrating lithium using a concentration process comprising counter-flow reverse osmosis operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic process diagram of a lithium recovery process according to one embodiment.

[0009] FIG. 2 is a schematic diagram of a vaporizer usable during a concentration stage of the process shown in FIG. 1

[0010] FIG. 3 is a schematic process diagram of a concentration stage of a lithium recovery process according to one embodiment.

[0011] FIG. 4 is a schematic process diagram of a portion of a concentration stage of a lithium recovery process according to another embodiment.

[0012] FIG. 5 is a schematic process diagram of an extraction stage according to one embodiment.

[0013] FIG. 6A-6C are schematic process diagrams of a lithium recovery process according to additional embodiments.

[0014] FIG. 7 is a schematic process diagram of a portion of a lithium recovery process.

DETAILED DESCRIPTION

[0015] Direct extraction of lithium is commonly used in lithium recovery from aqueous lithium sources. Some direct extraction processes employ a solid material to withdraw lithium selectively from a lithium source onto or into the withdrawal material. A recovery fluid is then contacted with the loaded withdrawal material to remove the lithium from the withdrawal material to form a lithium intermediate stream. The quantity of recovery fluid generally determines the concentration of lithium in the lithium intermediate stream, but unloading rate of ions from the withdrawal material can provide an effective upper limit to the concentration achievable.

[0016] In ion withdrawal, the withdrawal material is generally chosen to be selective to lithium. That may mean that many types of cations are removed from the source, but lithium is removed more readily than other cations. Thus, the ions removed by the withdrawal material include lithium possibly along with other impurities, such as monovalent cations sodium and potassium and divalent cations calcium and magnesium. Longer recovery processing can enhance ion removal in each cycle, but such measures are subject to diminishing returns as throughput declines. Thus, recovery fluid application rate is subject to an optimum which trades lithium intermediate stream concentration, recovery time, and degradation of loading capacity.

[0017] An ion withdrawal direct lithium extraction process may be an ion exchange or ion replacement process, where the withdrawal medium is pre-loaded with ions that are exchanged to the feed fluid while withdrawing other ions from the feed fluid. In such cases, the withdrawal and recovery processes are both typically ion exchange or ion replacement processes, but an ion exchange process where the recovery fluid does not replace ions can also be used, where ions are replaced for the withdrawal step by exposing the withdrawal medium to a third fluid for the purpose of preloading exchange ions.

[0018] Direct lithium extraction processes can also use a lithium selective electrochemical separation process. The lithium selective electrochemical separation process uses a voltage bias to

drive materials through a lithium selective membrane to separate lithium from an aqueous lithium source. The aqueous lithium source is brought into contact with a first side of the lithium selective membrane, and an aqueous eluent material is brought into contact with a second side of the lithium selective membrane, opposite from the first side. The voltage bias is applied within the aqueous lithium source and the aqueous eluent material to form an electric field within both materials and extending across the lithium selective membrane. The electric field provides a driving force to move, or increase movement of, charged species through the lithium selective membrane. The species motivated by the electric field to move through the lithium selective membrane depends on the configuration of the lithium selective membrane. For example, the lithium selective membrane may selectively pass lithium ions more than other ions or the lithium selective membrane may selective block passage of lithium ions more than other ions.

[0019] Direct lithium extraction processes that include lithium selective electrochemical separation processes use lithium selective membranes. Such membranes can include, or be made of, lithium selective materials such as lithium aluminum germanium phosphate, lithium aluminum titanium phosphate, lithium lanthanum titanates, or a metal organic framework type material such as UiO-66 with acid and amine groups. Such materials can be configured alone in a membrane structure or can be added to a support material, such as a resin, configured into a membrane structure.

[0020] In some cases, the direct lithium extraction process may be an adsorption process where ions are adsorbed from the aqueous lithium stream solution onto the surface of a solid adsorbent material that is selective to lithium, such as metal oxide, metal hydroxide or such material mixed with a resin. A desorbent solution is used to recover the withdrawn ions. In other cases, the direct lithium extraction process may be an absorption process where ions are absorbed from the brine solution into the bulk of a solid absorbent material that is selective to lithium. A desorbent solution is used in these cases, as well. These cases of pure sorption-desorption can require regeneration of the withdrawal medium because unloading of ions from medium is not quantitative.

[0021] FIG. 1 is a schematic process diagram of a lithium recovery process **100**, according to one embodiment. The process **100** uses an extraction stage **104** that performs direct lithium extraction, as described above. An extraction feed **105** is provided to the extraction stage **104** for withdrawal of ions, or electrochemical separation, to produce a lithium-depleted stream **107**. The extraction stage **104** is most effective where lithium concentration in the extraction feed **105** is at least about 70 ppm, for example at least about 100. Lithium sources having higher concentrations, for example as high as 1,000-3,000 ppm, of lithium can also be used.

[0022] In sorption/desorption embodiment where the adsorbent material is a solid, the adsorbent material may be stationary or fluidized within the vessel, or conveyed through one or more vessels or zones for contacting with the brine, for example in a counter-current format. In particular, the adsorbent material may be contained in a plurality of vessels in flow communication with one another and the vessels may be fluidly connected with a plurality of zones (ie inlets/outlets) during the extraction process. The extraction **104** may therefore take place continuously, for instance loading resin in a first vessel with lithium by fluidly connecting this vessel with the brine source while unloading resin in a second vessel by fluidly connecting the second vessel with the eluent and washing a third vessel using a strip solution. The extraction may be continuous counter-current adsorption desorption (CCAD). An example of a counter-current adsorption desorption that may be used is for instance described in U.S. Pat. No. 11,365,128 from EnergySource Minerals, which is hereby incorporated by reference as a description of an example process.

[0023] The lithium-depleted stream **107** may be separated into a reject stream and a fresh water stream using at least a membrane separation operation having a semi-permeable membrane, or a thermal vaporizer such as vaporizer **200** described below. The reject stream may be returned to the environment (i.e. reinjected into the geological formation) and the fresh water stream may be recycled into another stream of the process **100**. When a membrane separation operation is deployed it may be configured to perform electrodialysis, reverse osmosis, counter-flow reverse

osmosis, a combination of both reverse osmosis and counter-flow reverse osmosis such as described elsewhere herein. In that case, the lithium-depleted stream takes place of the lithium extract, the reject stream takes place of the lithium concentrate and the fresh water stream corresponds to the permeate stream.

[0024] The process **100** is configured to use dilute lithium sources. The sources can be salar brines, water leached from rock and clay formations and/or leaching resins, produced water from wells, mines, and geothermal installations, seawater, pre-treated aqueous lithium sources such as desalinators or electrochemical process effluents, recycled lithium from previous industrial applications, and other sources. Some of these sources can have very low concentrations of lithium (seawater, for example, averages about **0.4** ppm lithium) and very high relative concentrations of impurity ions such as sodium. Most of the impurity ions have solubility in water lower than lithium, so bringing concentration of lithium up in these materials generally precipitates solids, which must be managed.

[0025] The process **100** uses a feed preparation stage **102** to prepare a feed for the extraction stage **104**. The feed preparation stage **102** generally raises lithium concentration and TDS to an effective range for the extraction stage **104**. Any known way of raising concentration in an aqueous fluid may be used during feed preparation stage. For example, membrane processes, such as reverse osmosis (“RO”), nanofiltration (“NF,” sometimes also referred to as “loose” RO), and counter-flow reverse osmosis (such as CFRO® that is a counter-flow reverse osmosis product available from Gradient Corp. of Boston, Massachusetts, USA) can be used to remove water from an aqueous source. As another example, thermal methods using heat pumps, hot exhaust, solar radiation, and multi-effect evaporators can also be used. Evaporation, for example using cooling towers, atomizers, sprayers, and Carrier Gas Extraction® (also available from Gradient Corp.), mechanical or forced circulation evaporators or solar concentrators, can also be used to precipitate impurities and evaporate water. Combinations of such processes can also be used. In the disclosure, “concentration” is generally used for any water removal process, ie a process raising concentration of all species present in a solution equally.

[0026] Various modes of filtration can also be used in addition to the concentration to remove unwanted larger solids from brine sources, before and/or after concentrating the aqueous source. The feed preparation stage **102** may be configured to provide a lithium concentration ratio between the stream exiting the preparation stage **102** and the feed entering the preparation stage **102** of at least 10, or 20. The feed preparation stage **102** may be configured to operate in different ways based on the lithium source, so that the feed preparation stage **102** generally provides a lithium stream having at least about 70 ppm, for example about 100 ppm, such as about 150-200 ppm lithium ions, predominantly countered by chloride ions, for direct extraction in the extraction stage **104**, but higher concentrations of lithium could be used.

[0027] In an embodiment of the method, the feed preparation stage further comprises performing a purification process on the aqueous lithium source or the extraction feed to form a purified extraction feed. The feed preparation stage may include a purification process followed by a concentration process, or a concentration process followed by a purification process or a more complex combination, such as a first concentration process, a purification process and then a second concentration process. Each of the concentration processes may be as described in the disclosure, ie a membrane separation operation such as reverse osmosis, counter-flow reverse osmosis or both in combination, or an evaporation operation such as mechanical forced evaporation. The purification process may include one or more of the operations as described in relationship with the impurity stage, later in the specification, such as ion exchange, electrochemical process, solids removal, etc. In an embodiment the feed preparation stage includes a first concentration process using reverse osmosis, a purification process and a second concentration process using counter-flow reverse osmosis.

[0028] The direct extraction stage **104** generally operates as described above, either as an ion

withdrawal process or an electrochemical separation process, or a combination thereof.

[0029] In the ion withdrawal process, the withdrawal material is typically a material, for instance a resin, with a composition or surface preparation that increases affinity for lithium ions relative to other ions. The withdrawal material is typically housed in a vessel, and multiple vessels of withdrawal material are typically used so some resin beds can be regenerated without stopping production. Regeneration typically involves treatment with a hot fluid to achieve substantial removal of all ions from the resin. Where surface treatments or exchange ions are used, the regeneration can also include re-application of the surface treatment or exchange ions prior to placing the withdrawal bed back into service.

[0030] A recovery stream **109** is provided to the extraction stage to remove ions from the withdrawal material, thus forming a lithium intermediate **111**. The recovery stream **109** operates as an eluent that elutes lithium ions from the withdrawal medium into the recovery stream **109**. Flow rate of the recovery stream **109** is selected to concentrate lithium to a selected range, for instance of about 1,500 ppm to 3,000 ppm. Where low flow rate of the recovery stream **109** is used to achieve higher concentrations of lithium in the lithium intermediate **111**, concentration of impurity ions can also increase in the extraction stage **104**. The recovery stream **109** can be a water stream, which may be deionized, or a dilute brine stream having a low level of lithium ions, for example around 100 ppm. Selectivity for lithium ensures that impurity concentration rises less than lithium concentration.

[0031] Where the extraction stage **104** uses a lithium selective electrochemical separation process, the extraction feed **105** and the recovery stream **109** are brought into contact with opposite sides of a lithium selective membrane, as described above, and a voltage bias is applied across the membrane to perform direct extraction of lithium ions from the extraction feed. FIG. 5 is a schematic view of a lithium selective electrochemical separation cell **700**, according to one embodiment. The cell **700** has an enclosure **702** that is separated into a first volume **706** and a second volume **708** by a lithium selective membrane **704** disposed within the enclosure **702**. A lithium containing aqueous source is provided to the first volume **706** using a first volume inlet **710**. A receiving stream is provided to the second volume **708** using a second volume inlet **712**. A first electrode **714** is disposed to electrically couple to a material in the first volume **706**. For example, the first electrode **714** may be disposed within the first volume **706**, as shown in FIG. 5, or may be disposed on a wall of the first volume **306** or in a recess in fluid communication with the first volume **706**. A second electrode **316** is disposed to electrically couple to a material in the second volume **708** in any of the ways described above with respect to the first electrode **714**. A voltage bias is applied between the first electrode **714** and the second electrode **716** to provide an electric field between the first and second electrodes **714** and **716**, within the first and second volumes **706** and **708**, and across the membrane **704**. Depending on the configuration of the membrane **704**, the voltage bias may accelerate penetration of lithium ions through the membrane, while transport of other ions is either blocked or not accelerated as much as lithium ions. Alternately, the membrane **704** may be configured to block transport of lithium ions, so the electric field accelerates penetration of other ions relative to lithium. Application of the electric field thus enhances separation of lithium from other ions in the cell **700**.

[0032] The material in the first and second volumes **706** and **708** is changed by transport of ions across the membrane **704** using the driving force provided by the electric field raised by the electrodes **714** and **716**. The material of the first volume **306** is evacuated through a first volume outlet **720**, and the material of the second volume **308** is evacuated through a second volume outlet **722**. Depending on the configuration of the cell, the effluent from the first or second volumes **706** and **708** has increased lithium concentration relative to the corresponding feed material.

[0033] Where a membrane that is selectively permeable to lithium is used, current within the cell **700** is limited by lithium concentration in the first volume **706**. Concentrating lithium in the material provided to the first volume **706**, in such cases, for example up to a concentration of about

1,000 ppm or more, can increase separation rate and effectiveness. Reducing concentration of ions not transported by the membrane can also be useful to reduce buildup of material on the retentate side of the membrane, thus reducing the frequency of cleaning or maintenance operations. Effluent streams from RO desalination plants can be effectively treated using lithium selective electrochemical separation to recover lithium.

[0034] Where the cell **700** is used in the process **100**, the extraction feed **105** is provided to one of the inlets **710** and **712** and the recovery stream **109** is provided to the other. A membrane that is selectively permeable to lithium ions is used in the cell for the process **100**, such that lithium ions transport across the membrane from the extraction feed **105** to the recovery stream **109** while the extraction feed **105** is in contact with the first side of the membrane and the recovery stream **109** is in contact with the second, opposite, side of the membrane while the electric field enhances the ion transport.

[0035] Impurities are removed at an impurity stage **108**. Divalent impurities, such as calcium, and magnesium, are typically removed or reduced in the impurity stage **108** but any other impurity, for instance, transition metals, could be removed or reduced at that stage. Impurity stage is a stage where impurity are selectively removed or reduced, reducing the concentration of such impurities, while lithium concentration substantially remains the same. Any combination of membrane separation, ion exchange, electrostatic separation, and precipitation, any of which can be selective to at least one type of impurities by using components made to be selectively permeable, chemically affinitive, non-permeable, or chemically non-affinitive for impurities, can be used. Some such components are described elsewhere herein. Prior to removing impurities, concentration of the impurities can be increased to a level that optimizes the volume of water handled by the impurity stage. For most aqueous lithium streams, impurity removal is found to be most effective where impurity concentration is higher, so the lithium intermediate **111** can be treated in an impurity preparation stage **106** prior to treatment at the impurity stage **108**.

[0036] In the impurity preparation stage **106**, water is first removed from some or all of the lithium intermediate **111** to form an impurity stage feed **113** having elevated ion (i.e. lithium and impurity) concentration, along with a first removed stream **121**, which may be a water stream or brine stream. In other words, the impurity preparation stage concentrates the lithium intermediate. The impurity preparation stage **106** is generally operated to increase concentration of one or more impurities in the impurity stage feed to a value near the solubility limit of the one or more impurities in the impurity stage feed for example about 90% or 95% of the solubility limit. In an embodiment, the impurity preparation stage **106** is operated to raise a target divalent impurity, such as calcium, concentration in the impurity stage feed **113** to near the solubility limit in the eluent, for example about 90% or 95% of the solubility limit in the eluent. Removal of calcium, and other divalent ions, is then performed in the impurity stage **108** at near-optimal conditions to maximize removal of divalent impurities.

[0037] An example of impurity stage **108** includes solids filtration (for instance using clarification and/or filtering) for filtering the precipitated solids obtained during the impurity preparation stage **106**. The impurity stage **108** may include additional stages such selective removal of residual impurities, such as capture of divalent impurities such as calcium and magnesium using a ion exchange resin, precipitative methods, or an electrochemical process with an appropriate membrane to partition impurities. An ultrafiltration membrane that rejects divalent ions could also be deployed. The impurity stage may also include additional precipitation stage to selectively precipitate some impurities using a chemical treatment, such as coagulation-flocculation, followed by solids removal as described above. Coagulation-flocculation may include the addition of FeCl_3 with a base to form a floc of $\text{Fe}(\text{OH})_3$ used to remove suspended particles from the stream. The different treatments described herein as well as additional treatments for selectively removing any impurities may be implemented in any order. The impurity stage **108** results in a purified lithium stream **115** with high lithium concentration and very low concentration of

impurities, such as divalent impurities. The impurity preparation stage **106** is optional and the method may include a impurity stage **108** without impurity preparation stage **106**, in which case the lithium intermediate is routed to impurity stage **108**.

[0038] As previously described for the preparation stage **102**, any known way of raising concentration in an aqueous fluid may be used during impurity preparation stage **106**. For example, membrane processes, such as reverse osmosis (“RO”), nanofiltration (“NF,” sometimes also referred to as “loose” RO), and counter-flow reverse osmosis (CFRO® is a counter-flow reverse osmosis product available from Gradient Corp. of Boston, Massachusetts, USA) can be used to remove water from an aqueous source. As another example, thermal methods using heat pumps, hot exhaust, solar radiation, and multi-effect evaporators can also be used. Evaporation, for example using cooling towers, atomizers, sprayers, and Carrier Gas Extraction® (also available from Gradient Corp.), can also be used to precipitate impurities and evaporate water. Combinations of such processes can also be used. Various modes of filtration can also be used to remove unwanted larger solids from lithium sources. The impurity preparation stage **106** may be configured to provide a lithium concentration ratio between the stream exiting the impurity preparation stage **106** and the feed entering the impurity preparation stage **106** of at least **10**, or **20**. The impurity preparation stage **106** may be configured to operate differently with different compositions of the lithium intermediate stream **111**, to increase concentration of one or more impurities in the feed to the impurity stage **108** to a value near the solubility limit of the one or more impurities, as set forth above.

[0039] The purified lithium stream **115** that exits the impurity stage **108** can be routed to a post-extraction concentration stage **110**. The post-extraction concentration stage **110** raises the concentration of lithium by a factor of about 10.

[0040] The post-extraction concentration stage **110** produces a lithium concentrate **117** and one or more removed streams **119** that can be water streams, brine streams. The water and brine streams can be recycled to parts of the process **100** where ion concentrations are lower to facilitate processing. A portion of the lithium concentrate **117** can also be recycled and added to a lithium source to raise the lithium concentration of the lithium source and/or dilute impurities at the brine source. In an embodiment, the lithium concentrate **117** is routed to the conversion stage **112** that converts lithium chloride to lithium carbonate by adding sodium carbonate or directly to lithium hydroxide by electrochemical reaction.

[0041] In an embodiment, monovalent impurities can remain in the purified lithium stream **115**. In such cases, the post-extraction concentration stage **110** also concentrates any monovalent impurities that remains in the purified lithium stream **115**. Sodium and potassium have lower solubility limits in water than lithium, so concentrating the purified lithium stream **115** can precipitate sodium and potassium, which can be removed as solids in a solids removal stage taking place between post-extraction concentration **110** and conversion **112**. In other words, lithium concentrate **117** is filtered and filtered lithium concentrate is routed to lithium conversion. In an embodiment, the method may include a plurality of post-extraction concentration stages **110A**, **110B** can be a multi-operation concentrator with solids removal **114A**, **114B** between stages as shown on FIG. 7. A multi-operation concentrator with integrated solids removal may then be used. The first post-extraction concentration stage **110A** yields a first lithium concentrate **117A** and a first brine (or low TDS) stream **119A** that can be recycled elsewhere in the process. The first lithium concentrate **117A** is routed to the first solids removal **114A** that filters the precipitated solids within the second lithium concentrate as a first impurity slurry **149A**. The first filtered lithium concentrate **147A** is routed to the second post-extraction concentration stage **110B** and is further concentrated, yielding a second lithium concentrate **117B** and a second brine (or low TDS) stream **119B**. The second lithium concentrate **117B** is routed to the second solids removal **114B** that filters the precipitated solids within the second lithium concentrate as a second impurity slurry **149B**. The second filtered lithium concentrate **147A** is routed to conversion **112**. FIG. 7 exemplifies two concentration stages

followed by solids removal but any number of concentration stage with intermediate solids removal stage may be used. Each concentration operation can be configured to raise the concentration of lithium until monovalent impurity solids (such as sodium) become a burden. Solids can then be removed using any filtration process and further concentration can proceed. The aqueous lithium sources contemplated for use in the process **100** can have sodium concentration several orders of magnitude higher than lithium concentration, for example at least 100 or at least 1,000 times the lithium concentration, so the post-extraction concentration stage **110** and solids removal stage can be quite effective in removing large quantities of sodium with relative ease. Moreover, as lithium concentration increases, solubility of sodium ions declines, so each absolute increment of increased lithium concentration yields more sodium precipitation. Sodium overwhelmingly precipitates as chloride salt in the post-extraction concentration stage **110**, and solids can be removed by any suitable process including settling, centrifugation, vortex separation, and the like in the solids removal stage. At higher lithium concentrations, the most extreme filtration processes may become untenable due to the concentration of lithium ions, but processes such as evaporation and seeding can be used to remove water and precipitate sodium and potassium. The solids removal stage is an optional stage.

[0042] In one embodiment, a series of membrane separations is performed during the post-extraction concentration stage **110** to separate the lithium concentrate **117** with high lithium concentration, as a non-permeating stream, from a stream with low lithium concentration, as a permeating stream. The non-permeating stream, in this case, will also contain most impurities from the purified lithium stream **115**. A vaporizer may alternatively or additionally be used to carry out both post-extraction concentration stage **110** to further concentrate the lithium salt in the lithium concentrate **117** and solids removal stage **114**. The vaporizer yields a vaporizer water stream (equivalent to low TDS stream **119**), which can be recycled to parts of the process, as explained below, an impurity stream, which contains non-lithium cations such as sodium, potassium, magnesium, manganese, calcium, and the like (that is equivalent to the impurity slurry **129** of FIG. 7). The vaporizer also yields the filtered lithium concentrate **147**.

[0043] The concentration stage **110** produces a lithium concentrate **117** and one or more removed streams **119** that can be water streams, brine streams, and/or salt slurries. The water and brine streams can be recycled to parts of the process **100** where ion concentrations are lower to facilitate processing. A portion of the lithium concentrate **117** can also be recycled and added to a lithium source to raise the lithium concentration of the lithium source and/or dilute impurities at the brine source.

[0044] FIG. 2 is a schematic process diagram of a vaporizer **200** that can be used as, or as part of, the post-extraction concentration stage **110** and the solids removal stage **114**. The example vaporizer **200** allows concentration of lithium and removal of solids. The vaporizer **200** includes a vaporization vessel **202** that receives the purified lithium stream **115**. Heat is applied to the purified lithium stream **115** within the vaporization vessel **202** to vaporize water and concentrate lithium and other ions within the vessel **202**. A heater **204** is coupled to the vessel **202** to apply heat to the fluid within the vessel **202**. The heater **204** is shown here schematically as an element inserted into the interior of the vessel **202**, but heat input can be accomplished in any convenient manner.

[0045] The vessel **202** generally has a vaporization section **206** and a precipitation section **208**. Solids precipitate from the fluid as water is vaporized and solubility limits are reached. The vaporizer **200** is therefore also a precipitator of solids. Sodium precipitates as chloride, and potentially other salts due to trace amounts of other anions. Lithium generally remains in a concentrated solution, but some lithium salts can precipitate if enough water is removed by evaporation. Sodium solids generally settle below the lithium-rich solution due to density. The lithium solution is removed as the lithium concentrate **117**, which is removed from a lower part of the vaporization section **206**. Vaporized water is removed in an overhead stream **210** of the vaporization section **206**. Heat is recovered from the vaporized water by thermally contacting the

vaporized water with the purified lithium stream **115** in a heat exchanger **212**. The heated purified lithium stream **115** is provided to the vaporization section **206** of the vessel **202**, optionally using a valve or orifice to flash the heated purified lithium stream **115** within the vaporization section **206**. The vaporized water is at least partially condensed in the heat exchanger **212**, and a portion **250** of the vaporized water may be added to the lithium concentrate **117** to ensure all solids are dissolved prior to routing the lithium concentrate **117** to the conversion stage **112** of the process **100** (FIG. **1**). The remaining vaporized water exits as the vaporizer water stream **252**, which can be, or can be included in, one of the removed streams **119**.

[0046] Sodium solids, mainly chloride, along with other impurities such as calcium, potassium, magnesium, and manganese, also including any anion impurities, also precipitate in the vaporization section **206** of the vessel **202**, and due to higher density than the concentrated lithium solution settle into the precipitation section **208**. Note that the vaporization section **206** of the vessel **202** is sized to provide residence time for sodium precipitates to settle into the precipitation section **208**. A precipitate stream **214** is withdrawn from a lower portion of the precipitation section **208** and pumped to a settling vessel **216**. The sodium solids, along with other dense impurities, settle in the settling vessel **216** and are removed as an impurity stream **254**. Separated water or brine may be withdrawn from the settling vessel **216** and returned to the vaporization vessel **202** as a vaporization return stream **218** or recycled to any other location in the process (as explained below). Such vaporization return stream may be, or may be another part of, the removed stream **119**. In this case, the water or brine is returned at the bottom of the precipitation section **208** to fluidize solids that may collect at the bottom of the precipitation section **208**. The water or brine, or a portion thereof, can be returned to the vaporization vessel **202** at other points, or may be routed to other uses.

[0047] FIGS. **3** and **4** are schematic process diagrams showing embodiments of a concentration stage, for instance a post-extraction concentration stage **110**. In FIG. **3**, a concentration stage **300** includes two different membrane separation operations in series: a reverse osmosis operation **502** and a counter-flow reverse osmosis operation **504**. In this embodiment, the reverse osmosis operation **502** is upstream from the counter-flow reverse osmosis operation **504**. A lithium containing stream **301**, which can be the aqueous source, lithium intermediate **111** or the purified lithium stream **115** (FIG. **1**), is pressurized to a target pressure (preferably less than or equal to **2000** psi) using a pump **501** to yield a pressurized stream **503**, which is then provided to the reverse osmosis operation **502**.

[0048] The reverse osmosis operation **502** is represented as including a RO container **524** that has a semi-permeable membrane **526** disposed therein, which may be a lithium selective membrane. The semi-permeable membrane **526** may be a reverse osmosis membrane, a nanofiltration membrane or more generally any type of membrane that enables water molecules to permeate while lithium ions mostly do not permeate. The membrane **526** separates the RO container **524** into a first volume **528** that receives a stream to be concentrated, here the pressurized stream **503**, and a second volume **530** where permeating water molecules collect. The pressurized stream **503** enters the RO container **524** via an inlet **531** to the first volume **528**. The RO container **524** also has a first outlet **532** of the first volume **528** through which a preconcentrated stream **534** containing a higher concentration of lithium than the pressurized stream **503** or the lithium containing stream **301** exits the RO container **524**, and a second outlet **536** of the second volume **530** through which a dilute stream **538**, having a lower concentration of lithium, that passed through the membrane (i.e. a permeate stream) exits the RO container **524**.

[0049] The reverse osmosis operation **502** is represented in one stage with one container but it can also be in several stages, including a plurality of containers with identical or different semi-permeable membranes therein arranged in series. In such embodiments the preconcentrated stream **534** exiting a first RO container can be directed to the inlet of a second RO container to further concentrate the preconcentrated stream before a final preconcentrated stream is routed to the

counter-flow reverse osmosis operation. In other such cases the permeate stream **538** can be directed to the inlet of an additional RO to container to desalinate the permeate stream **538**. In still other embodiments, the reverse osmosis operation **502** may include a plurality of RO containers arranged in parallel.

[0050] FIG. **4** is a schematic process diagram of a reverse osmosis operation **400** that can be used as the reverse osmosis operation **502** (FIG. **3**). The reverse osmosis operation **400** has a first stage **601**, which is a parallel stage having two RO containers **602**, **604** (more than two can also be used) in parallel and a second stage **605** in series with the first stage **601** and having a RO container **606** (more than one RO container in series can also be used here) configured to receive a concentrated stream (i.e. a non-permeate stream) **603** of each RO container **602** and **604** as input. The RO container **606** of the second stage **605** further concentrates the concentrated streams **603** to yield the preconcentrated stream **534** (FIG. **3**). The reverse osmosis operation **400** is presented as an example of a way to arrange multiple RO containers in a reverse osmosis operation such as the reverse osmosis operation **502**. Using RO containers in series during reverse osmosis operation **502** can reduce the number of stages of the counter-flow reverse osmosis operation **504** as well as maximize permeate recovery, that can be re-used into the process as explained above, therefore reducing fresh water demand. Indeed, the permeate of each container in this case may be recycled to one or more stages of the process, in particular as eluent of the lithium extraction stage **102**.

[0051] Referring again to FIG. **3**, the counter-flow reverse osmosis operation **504** uses a plurality of n units **508.sub.1-508.sub.n**, in series, each unit comprising a semi-permeable membrane **510.sub.1-510.sub.n**, each of which may be a lithium selective membrane. Each semi-permeable membrane **510** may be a reverse osmosis membrane, a nanofiltration membrane or more generally any type of membrane that selectively enables water molecules to permeate while lithium ions mostly do not permeate. The units **508** may all have the same type of membrane or different types of membranes. Each membrane **510.sub.1-510.sub.n** separates each respective unit **508.sub.1-508.sub.n** into a first volume **512.sub.1-512.sub.n** to receive a stream to be concentrated, here a stream derived from the lithium containing stream **301**, and a second volume **514.sub.1-514.sub.n** to receive a permeating stream. The material that remains in the first volume **512** of each unit **508** is a non-permeating stream. Each unit comprises a first inlet **516.sub.1-516.sub.n** to receive the stream to be concentrated and a first outlet **518.sub.1-518.sub.n** to exit the concentrated stream (non-permeate stream) from the unit. Each unit **508** also has a second inlet **520.sub.1-520.sub.n** to receive the permeating stream of another sequential unit and a second outlet **522.sub.1-522n** to exit the permeating stream from the unit. Thus, for the units **508.sub.1-508.sub.n-1**, the non-permeating stream of the unit exits the first volume **512** of the unit and flows to the respective first volume **512** of the next unit in the series.

[0052] The preconcentrated stream **534** derived from the pressurized stream **503** (ultimately derived from a lithium bearing stream of the process **100**, such as the lithium intermediate **111** or the purified lithium stream **115**) and permeating streams of the units **508** flow in counter-current directions. That is, the non-permeating streams flow from unit **1** to unit n , generally through the first volumes **512**, while the permeating streams flow from unit n to unit **1**, generally through the second volumes **514**. The final non-permeating concentrated stream is collected at the first outlet **518.sub.n** of the n .sup.th unit **508.sub.n** and forms the lithium concentrate **117** (when using the operation **300** as the post-extraction concentration stage **110** of the process **100**). The final permeating stream is collected at the second outlet **522.sub.1** of the first unit **508.sub.1** and yields a dilute brine stream **535** that may be recycled into the counter-flow reverse osmosis operation and/or recycled to one or more other stages of the process **100**, for instance as a strip solution or as an eluent, for example with the recovery fluid **109**, in the extraction stage. Under some circumstances, the final permeating stream collected from the second volume of the first unit **508.sub.1** may be a fresh water stream having TDS less than about 2000 mg/l. The number of stages n may be between 2 and 10, optionally between 3 and 6 to limit the costs while concentrating the stream to a target

concentration. The concentration stage may also include a plurality of the counter-flow reverse osmosis operations **504** in parallel, each handling a portion of the flow to be concentrated.

[0053] In the embodiment shown in FIG. **3**, the concentration stage **300** first comprises the reverse osmosis operation **502**. The preconcentrated stream **534** is directed to the counter-flow reverse osmosis operation **504**, while the permeate stream **538** that has a low TDS (less than 2,000 mg/l, preferably less than 500 mg/l and preferably around 100 mg/l) can be recycled to another stream of the process **100**, for instance in the extraction stage **102** as eluent, for example with the recovery fluid **109**, or any other stage of the process **100** where fresh water can be used. During the counter-flow reverse osmosis operation **504**, the preconcentrated stream **534** passes through the n units **508.sub.1-508.sub.n** and the resulting final non-permeating stream is collected at the first outlet **518.sub.n** of the n th unit **508.sub.n**. The final non-permeating stream is depressurized, for example using a valve **540** or an orifice, and separated into a first portion that forms the lithium concentrate **117** (when used in the post-extraction concentration stage **110**) and a second portion that is sent back to the units **508.sub.1-508.sub.n** to flow through the second volumes **514** thereof along with permeating material. Using a lithium concentrate portion as a permeating stream in counter-flow reverse osmosis **504** increases operation efficiency. The depressurization may enable energy recovery by using the pressure of the final non-permeating stream exiting the n th first outlet **518 n** to drive, for example, a generator coupled with a turbine.

[0054] In each unit **508**, the lithium concentration of the non-permeating stream increases while the lithium concentration of the permeating stream decreases. That is, the non-permeating stream exiting unit m of the counter-flow reverse osmosis operation **504** at the respective first outlet **518 m** has higher lithium concentration than the non-permeating stream exiting unit $m-1$ at first outlet **518.sub. $m-1$** . Also, the permeating stream exiting unit m of the counter-flow reverse osmosis operation **504** at the respective second outlet **522.sub. m** has higher lithium concentration than the permeating stream exiting unit $m-1$ at second outlet **522.sub. $m-1$** . Said another way, lithium concentration increases, in the operation **504**, in streams flowing toward the n th unit and decreases in streams flowing toward the first unit. The dilute brine stream **535** may be recycled to the concentration stage **300** to recover any residual lithium in the dilute brine stream **535**, or to any preceding stage of the process **100**, as shown in FIG. **1**. In the concentration stage **300**, the dilute brine stream **535** can be mixed with the lithium containing stream **301**. In another embodiment, the concentration stage **100** may include only the counter-flow reverse osmosis operation **504** and no reverse osmosis operation **502** depending on lithium concentration in the lithium intermediate **111** and target concentration of the lithium concentrate **117**. In such case, a portion or all of the dilute brine stream **535** may be recycled to another stage of the process **100**.

[0055] A concentration stage including counter-flow reverse osmosis operation **504**, such as the concentration stage **300**, enables a concentration ratio between the stream exiting the n th unit and the stream entering the first unit, of 2 to 20. The concentration of the dilute brine **535** may be reduced to the point that a concentration ratio between the dilute stream entering the n th unit and the dilute stream exiting the first unit is 2 to 20. The stream entering the counter-flow reverse osmosis operation **504**, for example the preconcentrated stream **534**, has preferably a lithium concentration between 0.05% and 6% by weight, preferably between 0.5 and 3%. The lithium concentrate stream **117** at the exit of the counter-flow reverse osmosis operation **504** has a TDS (total dissolved solids) over 120,000 mg/l preferably over 200,000 mg/l and a lithium concentration over 2% by weight, preferably over 3.3% by weight. The dilute brine **535** at the second outlet **522.sub.1** of the first unit **508.sub.1** of the counter-flow reverse osmosis operation **504** has preferably a lithium concentration of less than 2% by weight, preferably less than 1.5% by weight. The counter-flow reverse osmosis operation **504** results in increased lithium concentration in the lithium concentrate **117**, compared to a more conventional method such as simple reverse osmosis operation, by 3 to 4 orders of magnitude, allowing recovery of more than 80%, preferably more than 90%, of the volume of the lithium intermediate **111** as the dilute brine stream **535**, when used

as the post-extraction concentration stage **110** of the process **100**. The counter-flow reverse osmosis operation **504** is an example of a second membrane separation operation that increases TDS to over 120,000 mg/l. However, a second membrane separation operation having different configuration and setup may also be used to reach such concentration, using for instance different equipment, or flow pattern, etc. Such operation is also covered by the current disclosure.

[0056] Combining a reverse osmosis operation **502** and a counter-flow reverse osmosis operation **504** limits the capital cost of the concentration stage by limiting the number of units in the counter-flow reverse osmosis operation. Furthermore, combining the operations **502** and **504** results in a dilute stream of the reverse osmosis operation **502** that can be recycled as an eluent in the lithium extraction operation, significantly reducing the fresh water needed in the extraction stage **104** (fresh water being mainly used as the recovery fluid **109**). In some cases, the dilute brine stream **535** can be recycled into a dilute brine target feed to one of the aqueous lithium source, the extraction feed **105**, the lithium intermediate **111**, the impurity stage feed, **113**, or the purified lithium stream **115**. The permeate stream **538** has a low lithium concentration and low TDS and is an efficient eluent whereas the dilute brine stream **535** may have a higher TDS that may not directly enable to elute lithium efficiently from the withdrawal material of the extraction stage **104** in some cases, or may pose higher barrier to diffusion of lithium through a membrane in an electrochemical process. In such cases the dilute brine stream **535** can be mixed with another stream having lower TDS, can be subjected to impurity removal before being used as eluent, or can otherwise be adjusted in composition for a target stream or operation of the process **100** to target a lithium concentration or ratio of lithium concentration to impurity concentration. A lower TDS stream, such as the permeate stream **538** can also be adjusted to target a lithium concentration or a ratio of lithium concentration to impurity concentration for a target stream or operation of the process **100**.

[0057] In one additional embodiment, the dilute brine stream **535** may be treated using a separate reverse osmosis operation independent from the concentration stage **300** (downstream of the counter-flow reverse osmosis operation **504**). In such cases, the reverse osmosis operation **502** can be optional. Any configuration or variation that concentrates lithium and yields fresh water (i.e. a water stream with TDS below 2,000 mg/l) that can be recycled elsewhere can be used as a post-extraction concentration stage **110**.

[0058] The configuration of FIGS. **3** and **4** described in relationship with the post-extraction concentration stage **110** may be also be used to increase lithium concentration as part of the feed preparation stage **102** or impurity removal preparation stage **106**. Where such a concentration configuration is used in the feed preparation stage **102**, the aqueous lithium source is the inlet stream to the RO container **524** and the feed for extraction is collected exiting the counter-flow reverse osmosis operation **504**. A permeate stream, such as the permeate stream **538** obtained from the reverse osmosis operation **502**, may also be recycled to other stages of the process **100** as described elsewhere in the application. When the feed preparation stage includes a first concentration process being reverse osmosis, a purification process and a second concentration process being counter-flow reverse osmosis, the feed preparation stage may be as described in relationship with FIGS. **3** & **4**, with an intermediate purification process on the preconcentrated stream to yield a purified preconcentrated stream that enters the counter-flow reverse osmosis in the volume **512.sub.1** of the first unit **508.sub.1**.

[0059] Where such configuration is used in the impurity preparation stage **106**, the lithium intermediate **111** is the inlet stream to the RO container **524** and the impurity stage feed **113** is collected exiting the counter-flow reverse osmosis operation **504**. The first removed stream **121** then corresponds to the permeate stream **538** obtained from the reverse osmosis operation **502** that may also be recycled to other stages of the process **100** as described elsewhere herein.

[0060] Referring again to FIG. **1**, multiple streams can be recycled in the process **100** to manage the concentration of lithium at each stage of the process **100** and to manage a ratio of lithium ions to impurity ions at each stage of the process, so that each stage of the process **100** can operate in an

optimal range. The composition of each stream transferred from one unit of the process **100** to another can be targeted to improve performance of the receiving unit. Thus, the impurity preparation stage **106** and the post-extraction concentration stage **110** produce respective removed streams **121** and **119**, which can be recycled to the recovery stream **109**. Each of the removed streams **121** and **119** can be water or dilute brine streams. The impurity stage **108** can also generate a water or dilute brine removed stream **123**, depending on the type of impurity reduction processes performed in the impurity stage **108**.

[0061] Each of the removed streams **119**, **121**, and **123**, or portions thereof, can, independently be returned to any stage of the process **100** to manage composition profile of lithium and impurity ions across the process **100** for optimal operation. All or part of any of the removed streams **119**, **121**, and **123** can, independently, be routed to the recovery stream **109** or to a first return **135** located in the lithium intermediate **111** between the extraction stage **104** and the impurity preparation stage **106**. All or part of any of the removed streams **119** and **123** can, independently, be routed to a second return located in the impurity stage feed **113** between the impurity preparation stage **106** and the impurity stage **108**. The removed stream **119**, or portion thereof, can be routed back to a third return located in the lithium intermediate stream **115** between the impurity stage **108** and the post-extraction concentration stage **110**. The removed streams are generally recycled backward in the process **100** to avoid sending streams with higher quantities of impurities forward in the process **100** toward the finished product end of the process **100**. Recycling the removed streams **119**, **121**, and **123** can manage water loading in the process **100** and minimize the need to make up water for processing.

[0062] Each of the intermediate streams **113**, **115**, and **117**, or portions thereof, can also be returned to any stage of the process **100** in an impurity feed recycle **125**, a concentrator feed recycle **127**, and a lithium concentrate recycle **129**, respectively. All or part of any of the recycle streams **125**, **127**, and **129** can be returned to a fourth return **133** located in the extraction feed **105** between the feed preparation stage **102** and the extraction stage **104**. All or part of any of the recycle streams **125**, **127**, and **129** can be returned to a fifth return **131** located in the lithium intermediate **111** between the extraction stage **104** and the impurity preparation stage **106**. All or part of any of the recycle streams **127** and **129** can be returned to a sixth return **141** located in the impurity feed stream **113** between the impurity preparation stage **106** and the impurity stage **108**. All of part of the recycle stream **129** can be returned to a seventh return **143** located in the concentrator feed **115** between the impurity stage **108** and the post-extraction concentration stage **110**.

[0063] The recycle streams **125**, **127**, and **129** have increasing lithium concentration and decreasing impurity concentration. The removed streams **119**, **123**, and **121** are generally water or dilute brine streams. These various streams are used to tune compositions at targeted locations in the process **100** to optimize performance of the individual units of the process **100**. For example, where a ratio of lithium to impurities in a particular process stream is lower or higher than optimal for performance of the unit immediately downstream, a low-or high-ion-concentration stream from a downstream location of the process can be selected to blend with the process stream to optimize the composition of the process stream.

[0064] The lithium concentrate **117** is routed to a conversion stage **112** to convert lithium chloride to lithium carbonate or lithium hydroxide, yielding a lithium product **145**. The conversion can be performed by known chemical, electrochemical, and hybrid processes. The lithium product **145** can be a lithium hydroxide product or a lithium carbonate product. The lithium product **145** can be a liquid solution of lithium hydroxide or lithium carbonate, a slurry of solid lithium hydroxide in a solution of lithium hydroxide, or a slurry of lithium carbonate in a solution of lithium carbonate.

[0065] FIG. **6A** is a schematic process diagram illustrating a process **800** according to one embodiment. The process **800** uses a concentration process **802**, and optionally a purification process **804**, in either order (with the purification process **804** shown in phantom conjoined with the concentration process **802** to illustrate that the processes can be performed in either order), as

the preparation stage **102**, a lithium selective electrochemical separation process **806** as the extraction stage **104**, and optionally any of the impurity stage **108**, and/or the impurity preparation stage **106**, and/or post-extraction concentration stage **110**, followed by the conversion stage **112**. These various stages are substantially as described above, and vaporization can be used in the post-extraction concentration stage **110**. The concentration process **802** can be a vaporization process, a filtration process, a membrane process, a CFRO process, a combination of a reverse osmosis and CFRO process or other suitable water-removal process as disclosed hereinabove. The purification process can be an ion-separation process, such as a selective membrane process or a chemical treatment, with solids removal.

[0066] FIG. **4B** is a schematic process diagram illustrating a process **820** according to another embodiment. The process **820** optionally uses concentration and purification processes **802** and **804**, in either order, as the optional preparation stage **102**, the lithium selective electrochemical separation process **806** as the extraction stage **104**, a CFRO process **808** optionally in combination with a reverse osmosis process as the concentration stage **110**, optional impurity removal stage **108** and optional impurity preparation stage **106** before the CFRO process **808**, followed by the conversion stage **112**.

[0067] FIG. **4C** is a schematic process diagram illustrating a process **850** according to another embodiment. The process **850** is like the process **820**, except that the extraction stage **104** is a withdrawal-type extraction stage and a lithium selective electrochemical separation process **852** is used for the impurity stage **108**, with a CFRO process **854** used as the impurity preparation stage **106**. The processes **4A-4C** illustrate the use of different separation processes together to recover lithium under different circumstances.

[0068] Methods of lithium recovery that use a preparation stage **102** and an extraction stage **104** can use an impurity stage **108** with or without the impurity preparation stage **106**, and with or without the post-extraction concentration stage **110**. Thus, the lithium intermediate **111** can be routed directly to the conversion stage **112**, or to the post-extraction concentration stage **110**, or to the impurity removal stage **108**. Methods described herein can have a preparation stage **102** and an extraction stage **104** directly followed by an impurity removal **108** without impurity preparation **106**, where the lithium intermediate **111** is routed directly to the impurity stage **108**. Other methods described herein can have a preparation stage **102** and an extraction stage **104** directly followed by a post-extraction concentration stage **110** without an impurity preparation stage **106** or an impurity stage **108**, where the lithium intermediate **111** is routed directly to the post-extraction concentration stage **110**). Methods described herein can also have an extraction stage **104**, an impurity preparation stage **106**, an impurity removal stage **108**, and a post-extraction concentration stage **110** followed by a conversion stage **112**.

[0069] The disclosure relates to a method of recovering lithium from a lithium source, comprising extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed. The input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. At least one of the concentration operations includes a counter-flow reverse osmosis operation. The method also includes generating a low TDS stream as a permeate from any of the one or more concentration operations, wherein the low TDS stream is recycled, ie directed to any operation of the method, especially having a fresh water need, or used as fresh water. The low TDS stream has a TDS under 2,000 mg/l, preferably under 500 mg/l.

[0070] One or more concentration operations may include a reverse osmosis operation upstream of a counter-flow reverse osmosis operation. The reverse osmosis separates the input stream into a preconcentrated stream and a permeate stream using a semi-permeable membrane, wherein the permeate stream is the low TDS stream.

[0071] The counter-flow reverse osmosis operation may include flowing the preconcentrated

stream into a plurality of reactors in series, each containing a semi-permeable membrane separating the reactor into a first volume and a second volume. The preconcentrated stream flows sequentially as a non-permeating stream into the first volume of each reactor and a permeating stream flows sequentially into the second volume of each reactor counter-current to the non-permeating stream. The non-permeating stream exiting the plurality of reactors yields a concentrated stream and the permeating stream exiting the plurality of reactors yields a dilute brine stream.

[0072] The dilute brine stream may be recycled into the reverse osmosis operation.

[0073] The at least one concentration operation may include pressurizing the input stream, especially before the counter-flow reverse osmosis operation, and preferably at a target pressure lower than membrane threshold pressure, in particular below 2000 psi. In an embodiment, the at least one concentration operation may include pressurizing the input stream before the reverse osmosis operation and depressurizing the permeating stream before flowing it into the second volume of the plurality of reactors.

[0074] The at least one concentration operation may include a feed preparation operation, wherein the input stream is an aqueous lithium source and the output feed is the extraction feed. The feed preparation operation takes place upstream of the extraction stage.

[0075] In an embodiment, the method includes feed performing a purification process on the aqueous lithium source or the extraction feed or a derivative thereof. The method may further comprise performing a purification process on the aqueous lithium source or the extraction feed to form a purified extraction feed, wherein extracting lithium from the extraction feed, or a derivative thereof, comprises extracting lithium from the purified extraction feed.

[0076] In an embodiment, the method comprises concentrating the aqueous source to yield a preconcentrated extraction feed in a first concentration process, such as a reverse osmosis process, performing the purification process on the preconcentrated extraction feed to yield a purified preconcentrated extraction feed, and concentrating the purified preconcentrated extraction feed using a second concentration process, such as a counter-flow reverse osmosis process, to yield the extraction feed. In this embodiment, the first and second concentration process may be any of the process described hereinabove, such as any membrane separation operation or evaporation operation. The purification process may be any combination of impurity precipitation, solids removal and divalent impurity selective removal.

[0077] The at least one concentration operation may include concentrating a stream derived from the lithium intermediate as the input stream to yield a lithium concentrate as the output feed. Such a concentration operation takes place downstream of the extraction stage, directly on the lithium intermediate or on a stream corresponding to the lithium intermediate that has undergone one or more additional operations. In an embodiment, the stream derived from the lithium intermediate is a first stream, and the method further comprises treating a second stream derived from the lithium intermediate in an impurity stage to remove impurities, and forming a purified lithium stream, wherein the first stream is the purified lithium stream. In other word there might be an impurity treatment operation (ie impurity stage) between the extraction stage and the concentration stage.

[0078] In such embodiment where there is an impurity stage, the one or more concentration operations may further include an impurity preparation stage. In such stage, the input stream is the lithium intermediate and the output feed is an an impurity stage feed, and the second stream (ie undergoing the impurity stage) is the impurity stage feed. In an embodiment, concentrating the lithium intermediate in the impurity preparation stage comprises increasing concentration of one or more impurities in the lithium intermediate to at least 90% of the solubility limit of the one or more impurities in the lithium intermediate

[0079] The method may further comprise converting lithium in the lithium concentrate to a lithium product. In particular, the lithium chloride in the lithium concentrate is converted to lithium carbonate and/or hydroxide. In an embodiment, the method includes routing the lithium concentrate or a derivative thereof to a solids removal stage, wherein the solids removal stage removes solids

from the lithium concentrate to yield a filtered lithium concentrate and routing the filtered lithium concentrate or a derivative thereof to conversion. In an embodiment, the method includes a first post-extraction concentration stage yielding a first lithium concentrate, a first solids removal stage removing solids from the first lithium concentrate or a derivative thereof to yield a first filtered lithium concentrate, a second post-extraction concentration stage concentrating the first filtered lithium concentrate or a derivative thereof to yield a second lithium concentrate and a solids removal stage removing solids from the second lithium concentrate or a derivative thereof to yield a second filtered lithium concentrate. The second filtered lithium concentrate or a stream derived thereof may be routed to conversion stage.

[0080] At least a portion of the low TDS stream is recycled into the extraction stage.

[0081] Extracting lithium from the extraction feed includes contacting the extraction feed with a lithium selective medium to load the medium with lithium and contacting an eluent stream with the lithium-loaded medium to form the lithium intermediate. low TDS stream may then be recycled into the eluent stream. Alternatively or additionally, the brine source stream yields a lithium depleted brine stream after having contacting the lithium selective medium, and using at least a membrane separation operation or thermal vaporizer to yield a reject stream and a fresh water stream, wherein the fresh water stream is recycled, ie directed to any operation of the method, especially having a fresh water need. The reject stream may be returned to the environment, ie reinjected in the geological formation.

[0082] The extraction stage may include a continuous counter-current adsorption desorption process.

[0083] In an embodiment, the extraction stage comprises a lithium selective electrochemical separation process.

[0084] The plurality of reactors may include 2 to 10, preferable 3 to 6, reactors. The lithium concentration range between the lithium concentrate and non-permeating stream may be between 2 and 20. The lithium concentration range between the lithium concentrate and dilute brine stream may be between 2 and 20. The TDS of the lithium concentration is over 120,000 mg/l and preferably over 200,000 mg/l. The lithium concentration of the lithium concentrate may be over 2.%, preferably over 3.3% weight.

[0085] At least a portion of the low TDS stream is recycled as a permeate target feed into another stream used in any stage of the method. The permeate target feed may for instance be one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0086] At least a portion of the dilute brine stream is recycled into a dilute brine target feed, and the dilute brine target feed is one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0087] In an embodiment, the at least one portion of the low TDS stream recycled into the permeate target feed and/or the at least one portion of the dilute brine stream recycled into the dilute brine target feed is adjusted to manage the concentration of lithium or to manage a ratio of lithium ions to impurity ions into the permeate, respectively dilute brine, target feed.

[0088] In an embodiment, the aqueous lithium source has a lithium concentration less than about 100 ppm, optionally less than 70 ppm, optionally less than 1 ppm.

[0089] In an embodiment, the aqueous lithium source has a total dissolved solids at least 1,000, optionally 10,000 times higher than the lithium concentration of said aqueous source.

[0090] The disclosure also relates to a method of recovering lithium from a lithium source, comprising extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. The at least one concentration operations includes concentrating a non-permeating

stream derived from the input stream to form the output feed using at least a membrane separation operation. The membrane separation operation include flowing the non-permeating stream in a plurality of reactors in series, wherein each reactor contains a semi-permeable membrane separating the reactor into a first and a second volumes, wherein the non-permeating stream flows into the first volume of the plurality of reactors. It also includes collecting the non-permeating stream at the outlet of the plurality of the reactors. A first portion of the non-permeating stream forms the output feed and a second portion of the non-permeating stream is recycled into the membrane separation operation as a permeating stream. It further includes flowing the permeating stream into a second volume of the plurality of reactors counter-current to the non-permeating stream.

[0091] In an embodiment, the membrane separation operation is a first membrane separation operation, and the at least one concentration operation includes a second membrane separation operation to concentrate the inlet stream upstream from the first membrane separation operation. The second membrane separation operation includes separating the inlet stream into a preconcentrated stream and a permeate stream using at least a semi-permeable membrane. The preconcentrated stream is the non-permeating stream of the first membrane separation operation.

[0092] The method may also include collecting the permeating stream at the outlet of the plurality of the reactors. The collected permeating stream forms a dilute brine stream.

[0093] The at least one concentration operation may also include pressurizing the non-permeating stream, especially before flowing it into the first volume of the plurality of reactors, preferably at a target pressure lower than membrane threshold pressure, in particular below 2000 psi, and depressurizing the permeating stream before flowing it into the second volume of the plurality of reactors.

[0094] The at least one concentration operation may also include a feed preparation operation that concentrates the aqueous lithium source to yield the extraction feed.

[0095] In an embodiment, the method includes performing a purification process on the aqueous lithium source or the extraction feed or a derivative thereof. The method may further comprise performing a purification process on the aqueous lithium source or the extraction feed to form a purified extraction feed, wherein extracting lithium from the extraction feed, or a derivative thereof, comprises extracting lithium from the purified extraction feed.

[0096] In an embodiment, the method comprises concentrating the aqueous source to yield a preconcentrated extraction feed in a first concentration process, such as a reverse osmosis process, performing the purification process on the preconcentrated extraction feed to yield a purified preconcentrated extraction feed, and concentrating the purified preconcentrated extraction feed using a second concentration process, such as a counter-flow reverse osmosis process, to yield the extraction feed. In this embodiment, the first and second concentration process may be any of the process described hereinabove, such as any membrane separation operation or evaporation operation. The purification process may be any combination of impurity precipitation, solids removal and divalent impurity selective removal.

[0097] The at least one concentration operation may include concentrating a stream derived from the lithium intermediate as the input stream to yield a lithium concentrate as the output feed. Such a concentration operation takes place downstream of the extraction stage, directly on the lithium intermediate or on a stream corresponding to the lithium intermediate that has undergone one or more additional operations. In an embodiment, the stream derived from the lithium intermediate is a first stream, and the method further comprises treating a second stream derived from the lithium intermediate in an impurity stage to remove impurities, and forming a purified lithium stream, wherein the first stream is the purified lithium stream. In other word there might be an impurity treatment operation (ie impurity stage) between the extraction stage and the concentration stage.

[0098] The impurity stage may include one or more of the following operations: impurity precipitation, solids removal and divalent impurity selective removal. The impurity precipitation

may comprise coagulation-flocculation. The divalent impurity selective removal may comprise an selective electrochemical separation process, that may include an impurity selective membrane, and/or a divalent impurity capture using ion exchange resin. In an embodiment, the impurity stage includes routing the stream derived from the lithium intermediate (ie lithium intermediate or a derivative thereof) in an impurity precipitation operation via coagulation-flocculation to yield a precipitate stream, routing the precipitate stream or a derivative thereof to solids removal to yield a filtered precipitate stream and a precipitate and routing the filtered precipitate stream or a derivative thereof to the divalent impurity selective removal to yield the purified stream. In another embodiment, the impurity stage includes solids removal only to remove impurities precipitated in the impurity preparation stage (detailed below). Other embodiments including one or more instance of one or more of the impurity precipitation, solids removal and divalent impurity selective removal in either order are also part of the current disclosure.

[0099] In such embodiment where there is an impurity stage, the one or more concentration operations may further include an impurity preparation stage. In such stage, the input stream is the lithium intermediate and the output feed is an an impurity stage feed, and the second stream (ie undergoing the impurity stage) is the impurity stage feed. In an embodiment, concentrating the lithium intermediate in the impurity preparation stage comprises increasing concentration of one or more impurities in the lithium intermediate to at least 90% of the solubility limit of the one or more impurities in the lithium intermediate. The impurity preparation stage may include a membrane separation operation, or evaporator. Any membrane separation (including counter-flow reverse osmosis, reverse osmosis or a combination thereof) may be used. Enhance or mechanical evaporators may be used as well. The impurity preparation stage is preferably configured so that the impurity stage feed has a TDS (total dissolved solids) over 120,000 mg/l preferably over 200,000 mg/l.

[0100] The method may further comprise converting lithium in the lithium concentrate to a lithium product. In particular, the lithium chloride in the lithium concentrate is converted to lithium carbonate and/or hydroxide. In an embodiment, the method includes routing the lithium concentrate or a derivative thereof to a solids removal stage, wherein the solids removal stage removes solids from the lithium concentrate to yield a filtered lithium concentrate and routing the filtered lithium concentrate or a derivative thereof to conversion. In an embodiment, the method includes a first post-extraction concentration stage yielding a first lithium concentrate, a first solids removal stage removing solids from the first lithium concentrate or a derivative thereof to yield a first filtered lithium concentrate, a second post-extraction concentration stage concentrating the first filtered lithium concentrate or a derivative thereof to yield a second lithium concentrate and a solids removal stage removing solids from the second lithium concentrate or a derivative thereof to yield a second filtered lithium concentrate. The second filtered lithium concentrate or a stream derived thereof may be routed to conversion stage.

[0101] At least a portion of the low TDS stream is recycled into a permeate target feed, and the permeate target feed may be used in any stage of the method, and may for instance correspond to one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0102] At least a portion of the dilute brine stream is recycled into a dilute brine target feed, and the dilute brine target feed is one one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0103] In an embodiment, the at least one portion of the low TDS stream recycled into the permeate target feed and/or the at least one portion of the dilute brine stream recycled into the dilute brine target feed is adjusted to manage the concentration of lithium or to manage a ratio of lithium ions to impurity ions into the permeate, respectively dilute brine, target feed.

[0104] At least a portion of the low TDS stream is recycled, preferably into the extraction stage.

[0105] Extracting lithium from the extraction feed includes contacting the extraction feed with a

lithium selective medium to load the medium with lithium and contacting an eluent stream with the lithium-loaded medium to form the lithium intermediate. low TDS stream may then be recycled into the eluent stream. Alternatively or additionally, the brine source stream yields a lithium depleted brine stream after having contacting the lithium selective medium, and using at least a membrane separation operation or thermal vaporizer to yield a reject stream and a fresh water stream, wherein the fresh water stream is recycled, ie directed to any operation of the method, especially having a fresh water need. The reject stream may be returned to the environment, ie reinjected in the geological formation.

[0106] The extraction stage may include a continuous counter-current adsorption desorption process.

[0107] In an embodiment, the extraction stage comprises a lithium selective electrochemical separation process.

[0108] The plurality of reactors may include 2 to 10, preferable 3 to 6, reactors. The lithium concentration range between the lithium concentrate and non-permeating stream may be between 2 and 20. The lithium concentration range between the lithium concentrate and dilute brine stream may be between 2 and 20. The TDS of the lithium concentration is over 120,000 mg/l and preferably over 200,000 mg/l. The lithium concentration of the lithium concentrate may be over 2.%, preferably over 3.3% weight.

[0109] In an embodiment, the aqueous lithium source has a lithium concentration less than about 100 ppm, optionally less than 70 ppm, optionally less than 1 ppm.

[0110] In an embodiment, the aqueous lithium source has a total dissolved solids at least 1,000, optionally 10,000 times higher than the lithium concentration of said aqueous source.

[0111] The disclosure also relates to a method of recovering lithium from a lithium source. The method comprises extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; and performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. The at least one concentration operations includes a first membrane separation operation, having a first semi-permeable membrane, yielding from the input stream a preconcentrated stream and a permeate stream, and a second membrane separation operation. The preconcentrated stream flows into a plurality of reactors in series, each containing a semi-permeable membrane separating the reactor into a first volume and a second volume, and the preconcentrated stream flows sequentially as a non-permeating stream into the first volume of each reactor. The non-permeating stream exiting the plurality of reactors yields the output stream. The second membrane operation yields a dilute brine stream that exits the second volume of at least one of the reactors, wherein the dilute brine stream is recycled into the first membrane separation operation.

[0112] The second membrane separation operation may include a permeating stream that flows sequentially into the second volume of the plurality of reactors, counter-current to the permeating stream, wherein at least a portion of the non-permeating stream is recycled into the permeating stream, wherein the permeating stream yields the dilute brine stream.

[0113] The The at least one concentration operations may also include pressurizing the non-permeating stream, especially before flowing it into the first volume of the plurality of reactors, preferably at a target pressure lower than membrane threshold pressure, in particular below 2000 psi, and depressurizing the permeating stream before flowing it into the second volume of the plurality of reactors.

[0114] The at least one concentration operation may also include a feed preparation operation that concentrates the aqueous lithium source to yield the extraction feed.

[0115] In an embodiment, the method includes performing a purification process on the aqueous lithium source or the extraction feed or a derivative thereof. The method may further comprise

performing a purification process on the aqueous lithium source or the extraction feed to form a purified extraction feed, wherein extracting lithium from the extraction feed, or a derivative thereof, comprises extracting lithium from the purified extraction feed.

[0116] In an embodiment, the method comprises concentrating the aqueous source to yield a preconcentrated extraction feed in a first concentration process, such as a reverse osmosis process, performing the purification process on the preconcentrated extraction feed to yield a purified preconcentrated extraction feed, and concentrating the purified preconcentrated extraction feed using a second concentration process, such as a counter-flow reverse osmosis process, to yield the extraction feed. In this embodiment, the first and second concentration process may be any of the process described hereinabove, such as any membrane separation operation or evaporation operation. The purification process may be any combination of impurity precipitation, solids removal and divalent impurity selective removal.

[0117] The at least one concentration operation may include concentrating a stream derived from the lithium intermediate as the input stream to yield a lithium concentrate as the output feed. Such a concentration operation takes place downstream of the extraction stage, directly on the lithium intermediate or on a stream corresponding to the lithium intermediate that has undergone one or more additional operations. In an embodiment, the stream derived from the lithium intermediate is a first stream, and the method further comprises treating a second stream derived from the lithium intermediate in an impurity stage to remove impurities, and forming a purified lithium stream, wherein the first stream is the purified lithium stream. In other word there might be an impurity treatment operation (ie impurity stage) between the extraction stage and the concentration stage.

[0118] The impurity stage may include one or more of the following operations: impurity precipitation, solids removal and divalent impurity selective removal. The impurity precipitation may comprise coagulation-flocculation. The divalent impurity selective removal may comprise an selective electrochemical separation process, that may include an impurity selective membrane, and/or a divalent impurity capture using ion exchange resin. In an embodiment, the impurity stage includes routing the stream derived from the lithium intermediate (ie lithium intermediate or a derivative thereof) in an impurity precipitation operation via coagulation-flocculation to yield a precipitate stream, routing the precipitate stream or a derivative thereof to solids removal to yield a filtered precipitate stream and a precipitate and routing the filtered precipitate stream or a derivative thereof to the divalent impurity selective removal to yield the purified stream. In another embodiment, the impurity stage includes solids removal only to remove impurities precipitated in the impurity preparation stage (detailed below). Other embodiments including one or more instance of one or more of the impurity precipitation, solids removal and divalent impurity selective removal in either order are also part of the current disclosure.

[0119] In such embodiment where there is an impurity stage, the one or more concentration operations may further include an impurity preparation stage. In such stage, the input stream is the lithium intermediate and the output feed is an an impurity stage feed, and the second stream (ie undergoing the impurity stage) is the impurity stage feed. In an embodiment, concentrating the lithium intermediate in the impurity preparation stage comprises increasing concentration of one or more impurities in the lithium intermediate to at least 90% of the solubility limit of the one or more impurities in the lithium intermediate

[0120] At least a portion of the permeate stream may be recycled into is a permeate target feed. The permeate target feed may be any stream used in one or more stages of the method, for instance one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0121] At least a portion of the permeate stream may be recycled into the extraction stage.

[0122] Extracting lithium from the extraction feed includes contacting the extraction feed with a lithium selective medium to load the medium with lithium and contacting an eluent stream with the lithium-loaded medium to form the lithium intermediate. low TDS stream may then be recycled

into the eluent stream. Alternatively or additionally, the brine source stream yields a lithium depleted brine stream after having contacting the lithium selective medium, and using at least a membrane separation operation or thermal vaporizer to yield a reject stream and a fresh water stream, wherein the fresh water stream is recycled, ie directed to any operation of the method, especially having a fresh water need. The reject stream may be returned to the environment, ie reinjected in the geological formation.

[0123] The extraction stage may include a continuous counter-current adsorption desorption process.

[0124] In an embodiment, the extraction stage comprises a lithium selective electrochemical separation process.

[0125] The plurality of reactors may include 2 to 10, preferable 3 to 6, reactors. The lithium concentration range between the lithium concentrate and non-permeating stream may be between 2 and 20. The lithium concentration range between the lithium concentrate and dilute brine stream may be between 2 and 20. The TDS of the lithium concentration is over 120,000 mg/l and preferably over 200,000 mg/l. The lithium concentration of the lithium concentrate may be over 2 %, preferably over 3.3% weight.

[0126] In an embodiment, the aqueous lithium source has a lithium concentration less than about 100 ppm, optionally less than 70 ppm, optionally less than 1 ppm.

[0127] In an embodiment, the aqueous lithium source has a total dissolved solids at least 1,000, optionally 10,000 times higher than the lithium concentration of said aqueous source.

[0128] The disclosure also relates to a method of recovering lithium from a brine source. The method comprises extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; and performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. The at least one concentration operations may include a first membrane separation operation, having a first semi-permeable membrane, yielding from the input stream a preconcentrated stream and a permeate stream, and concentrating the preconcentrated stream using a second membrane separation operation, wherein the second membrane operation includes a plurality of reactors in series each having a semi-permeable membrane to yield the output stream, wherein the second membrane separation operation is configured so that the lithium concentrate has a TDS over 120,000 mg/l, preferably over 200,000 mg/l.

[0129] In an embodiment, in the second membrane operation, each semi-permeable membrane separate the associated reactor into a first volume and a second volume, wherein the preconcentrated stream flows sequentially as a non-permeating stream into the first volume of each reactor, and wherein the second membrane operation yields a dilute brine stream that exits the second volume of at least one of the reactors, wherein the dilute brine stream is recycled into the first membrane separation operation.

[0130] The second membrane separation operation may include a permeating stream that flows sequentially into the second volume of the plurality of reactors, counter-current to the permeating stream, wherein at least a portion of the non-permeating stream is recycled into the permeating stream, wherein the permeating stream yields the dilute brine stream.

[0131] The at least one concentration operations may also include pressurizing the non-permeating stream, especially before flowing it into the first volume of the plurality of reactors, preferably at a target pressure lower than membrane threshold pressure, in particular below 2000 psi, and depressurizing the permeating stream before flowing it into the second volume of the plurality of reactors.

[0132] The at least one concentration operation may also include a feed preparation operation that concentrates the aqueous lithium source to yield the extraction feed.

[0133] In an embodiment, the method includes feed performing a purification process on the

aqueous lithium source or the extraction feed or a derivative thereof. The method may further comprise performing a purification process on the aqueous lithium source or the extraction feed to form a purified extraction feed, wherein extracting lithium from the extraction feed, or a derivative thereof, comprises extracting lithium from the purified extraction feed.

[0134] In an embodiment, the method comprises concentrating the aqueous source to yield a preconcentrated extraction feed in a first concentration process, such as a reverse osmosis process, performing the purification process on the preconcentrated extraction feed to yield a purified preconcentrated extraction feed, and concentrating the purified preconcentrated extraction feed using a second concentration process, such as a counter-flow reverse osmosis process, to yield the extraction feed. In this embodiment, the first and second concentration process may be any of the process described hereinabove, such as any membrane separation operation or evaporation operation. The purification process may be any combination of impurity precipitation, solids removal and divalent impurity selective removal.

[0135] The at least one concentration operation may include concentrating a stream derived from the lithium intermediate as the input stream to yield a lithium concentrate as the output feed. Such a concentration operation takes place downstream of the extraction stage, directly on the lithium intermediate or on a stream corresponding to the lithium intermediate that has undergone one or more additional operations. In an embodiment, the stream derived from the lithium intermediate is a first stream, and the method further comprises treating a second stream derived from the lithium intermediate in an impurity stage to remove impurities, and forming a purified lithium stream, wherein the first stream is the purified lithium stream. In other word there might be an impurity treatment operation (ie impurity stage) between the extraction stage and the concentration stage.

[0136] The impurity stage may include one or more of the following operations: impurity precipitation, solids removal and divalent impurity selective removal. The impurity precipitation may comprise coagulation-flocculation. The divalent impurity selective removal may comprise an selective electrochemical separation process, that may include an impurity selective membrane, and/or a divalent impurity capture using ion exchange resin. In an embodiment, the impurity stage includes routing the stream derived from the lithium intermediate (ie lithium intermediate or a derivative thereof) in an impurity precipitation operation via coagulation-flocculation to yield a precipitate stream, routing the precipitate stream or a derivative thereof to solids removal to yield a filtered precipitate stream and a precipitate and routing the filtered precipitate stream or a derivative thereof to the divalent impurity selective removal to yield the purified stream. In another embodiment, the impurity stage includes solids removal only to remove impurities precipitated in the impurity preparation stage (detailed below). Other embodiments including one or more instance of one or more of the impurity precipitation, solids removal and divalent impurity selective removal in either order are also part of the current disclosure.

[0137] In such embodiment where there is an impurity stage, the one or more concentration operations may further include an impurity preparation stage. In such stage, the input stream is the lithium intermediate and the output feed is an an impurity stage feed, and the second stream (ie undergoing the impurity stage) is the impurity stage feed. In an embodiment, concentrating the lithium intermediate in the impurity preparation stage comprises increasing concentration of one or more impurities in the lithium intermediate to at least 90% of the solubility limit of the one or more impurities in the lithium intermediate

[0138] At least a portion of the permeate stream may be recycled into is a permeate target feed. The permeate target feed may be used in any stage of the method and for instance may be one of the aqueous lithium source, extraction feed, lithium intermediate, impurity stage feed or purified lithium stream.

[0139] At least a portion of the permeate stream may be recycled into the extraction stage.

[0140] Extracting lithium from the extraction feed includes contacting the extraction feed with a lithium selective medium to load the medium with lithium and contacting an eluent stream with the

lithium-loaded medium to form the lithium intermediate. low TDS stream may then be recycled into the eluent stream. Alternatively or additionally, the brine source stream yields a lithium depleted brine stream after having contacting the lithium selective medium, and using at least a membrane separation operation or thermal vaporizer to yield a reject stream and a fresh water stream, wherein the fresh water stream is recycled, ie directed to any operation of the method, especially having a fresh water need. The reject stream may be returned to the environment, ie reinjected in the geological formation.

[0141] The extraction stage may include a continuous counter-current adsorption desorption process.

[0142] In an embodiment, the extraction stage comprises a lithium selective electrochemical separation process.

[0143] The plurality of reactors may include 2 to 10, preferable 3 to 6, reactors. The lithium concentration range between the lithium concentrate and non-permeating stream may be between 2 and 20. The lithium concentration range between the lithium concentrate and dilute brine stream may be between 2 and 20. The TDS of the lithium concentration is over 120,000 mg/l and preferably over 200,000 mg/l. The lithium concentration of the lithium concentrate may be over 2 %, preferably over 3.3% weight.

[0144] In an embodiment, the aqueous lithium source has a lithium concentration less than about 100 ppm, optionally less than 70 ppm, optionally less than 1 ppm.

[0145] In an embodiment, the aqueous lithium source has a total dissolved solids at least 1,000, optionally 10,000 times higher than the lithium concentration of said aqueous source.

[0146] The disclosure also relates to a method of recovering lithium from a brine source, comprising extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate. The method also includes performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed. The at least one concentration operations includes at least a membrane separation operation, wherein at least one the membrane separation operation includes a plurality of reactors in series each having a semi-permeable membrane, yields a lithium concentrate and a dilute brine stream, and is configured so that the lithium concentrate has a TDS over 120,000 mg/l, preferably over 200,000 mg/l. The method also includes separating the dilute brine stream using a semi-permeable membrane into two streams including a permeate stream, wherein the permeate stream has a TDS under 2,000 mg/l, preferably under 500 mg/l, and recycling the permeate stream, ie directing it to any operation of the method, especially having a fresh water need.

[0147] In an embodiment, the one or more concentration operations include a first membrane separation operation yielding a preconcentrated stream and a diluted stream from the input stream, a second membrane operation, wherein the at least one membrane separation is the second membrane operation. The method also includes providing the dilute brine stream into the first membrane separation operation, wherein the diluted stream is the permeate stream.

[0148] In an embodiment, in the second membrane separation operation, each semi-permeable membrane separate the associated reactor into a first volume and a second volume, wherein a stream derived from the input stream flows sequentially as a non-permeating stream into the first volume of each reactor, and wherein the dilute brine stream that exits the second volume of at least one of the reactors, wherein the dilute brine stream is recycled into the first membrane separation operation. The second membrane separation operation may also have a permeating stream that flows sequentially into the second volume of the plurality of reactors, counter-current to the permeating stream, wherein at least a portion of the non-permeating stream is recycled into the permeating stream, wherein the permeating stream yields the dilute brine stream.

[0149] The at least one concentration operation may include a feed preparation operation that

concentrates the aqueous lithium source to yield the extraction feed.

[0150] The at least one concentration operation may include concentrating a stream derived from the lithium intermediate to yield a lithium concentrate. Such a concentration operation takes place downstream of the extraction stage, directly on the lithium intermediate or on a stream corresponding to the lithium intermediate that has undergone one or more additional operations. In an embodiment, the stream derived from the lithium intermediate is a first stream, and the method further comprises treating a second stream derived from the lithium intermediate in an impurity stage to remove impurities, and forming a purified lithium stream, wherein the first stream is the purified lithium stream. In other word there might be an impurity treatment operation (ie impurity stage) between the extraction stage and the concentration stage.

[0151] The impurity stage may include one or more of the following operations: impurity precipitation, solids removal and divalent impurity selective removal. The impurity precipitation may comprise coagulation-flocculation. The divalent impurity selective removal may comprise an selective electrochemical separation process, that may include an impurity selective membrane, and/or a divalent impurity capture using ion exchange resin. In an embodiment, the impurity stage includes routing the stream derived from the lithium intermediate (ie lithium intermediate or a derivative thereof) in an impurity precipitation operation via coagulation-flocculation to yield a precipitate stream, routing the precipitate stream or a derivative thereof to solids removal to yield a filtered precipitate stream and a precipitate and routing the filtered precipitate stream or a derivative thereof to the divalent impurity selective removal to yield the purified stream. In another embodiment, the impurity stage includes solids removal only to remove impurities precipitated in the impurity preparation stage (detailed below). Other embodiments including one or more instance of one or more of the impurity precipitation, solids removal and divalent impurity selective removal in either order are also part of the current disclosure.

[0152] In such embodiment where there is an impurity stage, the one or more concentration operations may further include an impurity preparation stage. In such stage, the input stream is the lithium intermediate and the output feed is an an impurity stage feed, and the second stream (ie undergoing the impurity stage) is the impurity stage feed. In an embodiment, concentrating the lithium intermediate in the impurity preparation stage comprises increasing concentration of one or more impurities in the lithium intermediate to at least 90% of the solubility limit of the one or more impurities in the lithium intermediate

[0153] At least a portion of the permeate stream may recycled into the extraction stage.

[0154] In an embodiment, extracting lithium from the extraction feed includes contacting the extraction feed stream with a lithium selective medium to load the medium with lithium and contacting an eluent stream with the lithium-loaded medium to form the lithium intermediate. The permeate stream may be recycled into the eluent stream. Additionnally or alternatively, the brine source stream yields a lithium depleted brine stream after having contacting the lithium selective medium, and using at least a membrane separation operation or thermal vaporizer to yield a reject stream and a fresh water stream, wherein the fresh water stream is recycled, ie directed to any operation of the method, especially having a fresh water need.

[0155] The extraction stage may include a continuous counter-current adsorption desorption process.

[0156] The plurality of reactors may include 2 to 10, preferable 3 to 6, reactors. The lithium concentration range between the lithium concentrate and non-permeating stream may be between 2 and 20. The lithium concentration range between the lithium concentrate and dilute brine stream may be between 2 and 20. The lithium concentration of the lithium concentrate may be over 2.%, preferably over 3.3% weight.

[0157] The method may also include pressurizing the input stream, especially before the at least one membrane separation operation, preferably at a target pressure lower than membrane threshold pressure, in particular below 2000 psi.

[0158] In the above, when reading “a first stream derived from a second stream”, “a first stream obtained from a second stream” or ‘a first stream derivative of a second stream’ shall be interpreted either as the first stream being the second stream or resulting from one or more operations performed on the second stream to change its properties such as its composition.

[0159] While the foregoing is directed to embodiments of the present invention, other and further embodiments of the present disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of recovering lithium from a lithium source, comprising: extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed; wherein at least one of the concentration operations includes a counter-flow reverse osmosis operation, generating a low TDS stream as a permeate from any of the one or more concentration operations, wherein the low TDS stream is recycled or used as fresh water.
2. The method of claim 1, wherein one or more concentration operations include a reverse osmosis operation upstream of a counter-flow reverse osmosis operation, wherein the reverse osmosis separates the input stream into a preconcentrated stream and a permeate stream using a semi-permeable membrane, wherein the permeate stream is the low TDS stream.
3. The method of claim 1, wherein the counter-flow reverse osmosis operation includes flowing the preconcentrated stream into a plurality of reactors in series, each containing a semi-permeable membrane separating the reactor into a first volume and a second volume, wherein the preconcentrated stream flows as a non-permeating stream sequentially into the first volume of each reactor and a permeating stream flows sequentially into the second volume of each reactor counter-current to the non-permeating stream, wherein the non-permeating stream exiting the plurality of reactors yields a concentrated stream and the permeating stream exiting the plurality of reactors yields a dilute brine stream.
4. The method of claim 3, wherein the dilute brine stream is recycled into the reverse osmosis operation.
5. The method of claim 1, wherein the at least one concentration operation includes a feed preparation operation, wherein the input stream is an aqueous lithium source and the output feed is the extraction feed.
6. The method of claim 1, wherein the at least one concentration operation includes concentrating a stream derived from the lithium intermediate as the input stream to yield a lithium concentrate as the output feed.
7. The method of claim 1, wherein at least a portion of the low TDS stream is recycled into the extraction stage.
8. The method of claim 1, wherein extracting lithium from the extraction feed includes contacting the extraction feed with a lithium selective medium to load the medium with lithium and contacting an eluent stream with the lithium-loaded medium to form the lithium intermediate, wherein the low TDS stream is recycled into the eluent stream.
9. A method of recovering lithium from a lithium source, comprising: extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed; wherein the at least one concentration operations includes a first membrane separation operation, having a first semi-

permeable membrane, yielding from the input stream a preconcentrated stream and a permeate stream, a second membrane separation operation, wherein the preconcentrated stream flows into a plurality of reactors in series, each containing a semi-permeable membrane separating the reactor into a first volume and a second volume, wherein the preconcentrated stream flows sequentially as a non-permeating stream into the first volume of each reactor, wherein the non-permeating stream exiting the plurality of reactors yields the output stream, and wherein the second membrane separation operation yields a dilute brine stream that exits the second volume of at least one of the reactors, wherein the dilute brine stream is recycled into the first membrane separation operation.

10. The method of claim 9, wherein the second membrane separation operation includes a permeating stream that flows sequentially into the second volume of the plurality of reactors, counter-current to the permeating stream, wherein at least a portion of the non-permeating stream is recycled into the permeating stream, wherein the permeating stream yields the dilute brine stream.

11. The method of claim 9, wherein the at least one concentration operation include a feed preparation operation that concentrates the aqueous lithium source to yield the extraction feed.

12. The method of claim 9, wherein the at least one concentration operation includes concentrating a stream derived from the lithium intermediate to yield a lithium concentrate.

13. The method of claim 9, wherein at least a portion of the permeate stream is recycled into the extraction stage.

14. A method of recovering lithium from a brine source, comprising: extracting lithium from an extraction feed using direct lithium extraction in an extraction stage to yield a lithium intermediate; performing one or more concentration operations, each concentration operation concentrating an input stream to yield an output feed, wherein the input stream is obtained from the lithium intermediate and/or the extraction feed is obtained from the output feed; wherein the at least one concentration operations includes at least a membrane separation operation, wherein at least one the membrane separation operation includes a plurality of reactors in series each having a semi-permeable membrane, yields a lithium concentrate and a dilute brine stream, and is configured so that the lithium concentrate has a TDS over 120,000 mg/l, preferably over 200,000 mg/l, separating the dilute brine stream using a semi-permeable membrane into two streams including a permeate stream, wherein the permeate stream has a TDS under 2,000 mg/l, preferably under 500 mg/l, recycling the permeate stream.

15. The method of claim 14, wherein the one or more concentration operations include: a first membrane separation operation yielding a preconcentrated stream and a diluted stream from the input stream, a second membrane operation, wherein the at least one membrane separation is the second membrane operation, and the method includes providing the dilute brine stream into the first membrane separation operation, wherein the diluted stream is the permeate stream.

16. The method of claim 14, wherein, in the second membrane separation operation, each semi-permeable membrane separate the associated reactor into a first volume and a second volume, wherein a stream derived from the input stream flows sequentially as a non-permeating stream into the first volume of each reactor, and wherein the dilute brine stream that exits the second volume of at least one of the reactors, wherein the dilute brine stream is recycled into the first membrane separation operation.

17. The method of claim 16, wherein the second membrane separation operation includes a permeating stream that flows sequentially into the second volume of the plurality of reactors, counter-current to the permeating stream, wherein at least a portion of the non-permeating stream is recycled into the permeating stream, wherein the permeating stream yields the dilute brine stream.

18. The method of claim 14, wherein the at least one concentration operation include a feed preparation operation that concentrates the aqueous lithium source to yield the extraction feed.

19. The method of claim 14, wherein the at least one concentration operation includes concentrating a stream derived from the lithium intermediate to yield a lithium concentrate.

20. The method of claim 14, wherein at least a portion of the permeate stream is recycled into the extraction stage.
